

Agentic Data Quality Guardian - CoCoQuest 2026

01-setup-success.png - 1

My Workspace > Untitled 2.sql

▶ ▼

ACCOUNTADMIN • COMPUTE_WH (X-Small) Choose database Share ...

1

SELECT COUNT(*) FROM DQ_GUARDIAN.RAW.CUSTOMERS; -- expect 5100

2

SELECT COUNT(*) FROM DQ_GUARDIAN.RAW.ORDERS; -- expect 12000

3

4

◆ SELECT AI_COMPLETE('llama3.1-70b', 'Reply with exactly: Cortex is working');

Results (just now)

▼

Table Chart Pivot

🔍 📄 1 row ⓘ 31ms 🔍 ⬇ ⌚

	# COUNT(*)	
1	5100	

01-setup-success.png - 2

My Workspace > Untitled 2.sql

▶ ▼

ACCOUNTADMIN • COMPUTE_WH (X-Small) Choose database Share ...

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SELECT COUNT(*) FROM DQ_GUARDIAN.RAW.CUSTOMERS; -- expect 5100

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◆ SELECT AI_COMPLETE('llama3.1-70b', 'Reply with exactly: Cortex is working');

◆ Add to Chat

Explain

Quick edit

Format

Results (just now)

▼

Table Chart Pivot

🔍 📄 1 row ⓘ 47ms 🔍 ⬇ ⌚

	# COUNT(*)	
1	12000	

02-cortex-check.png

The screenshot shows a Jupyter Notebook interface with a dark theme. The top bar includes tabs for 'RAG.sql', 'accelerate-app-dev-cortex-code.ipynb', 'Untitled.sql', 'Untitled 1.sql', 'Untitled 2.sql', and 'streamlit_app.py'. The 'Untitled 2.sql' tab is active, displaying the following SQL queries:

```
1 SELECT COUNT(*) FROM DQ_GUARDIAN.RAW.CUSTOMERS; -- expect 5100
2 SELECT COUNT(*) FROM DQ_GUARDIAN.RAW.ORDERS; -- expect 12000
3
4 SELECT AI_COMPLETE('llama3.1-70b', 'Reply with exactly: Cortex is working');
```

Below the code editor, there are buttons for 'Add to Chat', 'Explain', 'Quick edit', and 'Format'. The 'Results (just now)' section shows a table with one row:

1	"Cortex is working"

At the bottom, there are icons for search, filter, and other table operations. The text '1 row' and '895ms' are displayed next to the table.

03-app-home.png

The screenshot shows a Streamlit application titled 'Agentic Data Quality Guardian'. The main heading is 'Data health dashboard'. Below this, there are four metrics:

- Source health: 81.1
- Sandbox (fixed) health: 84.3 (with a green arrow and '3.2' indicating an increase)
- Scans run: 2
- Failing rules (source): 18 / 36

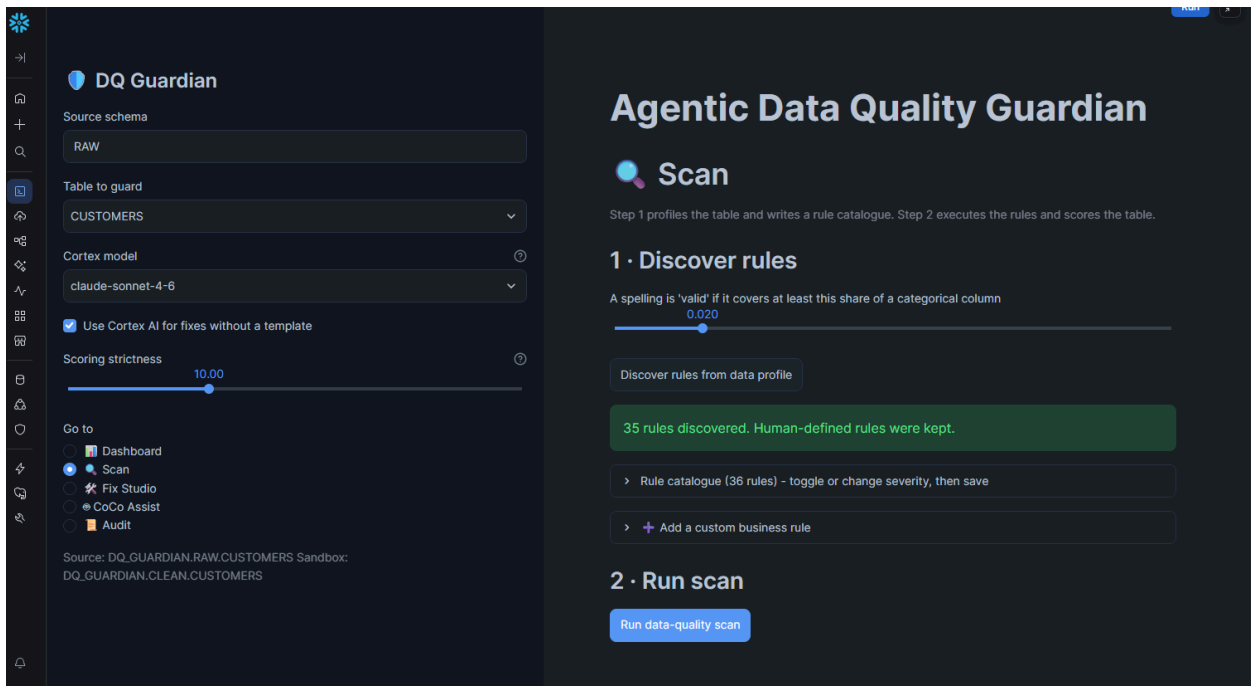
Below the metrics, there is a bar chart titled 'Quality by dimension' showing five dimensions: ACCURACY, COMPLETENESS, CONSISTENCY, UNIQUENESS, and VALIDITY. Each dimension has a blue bar and a yellow bar. The y-axis ranges from 0 to 200.

On the right side, there is a chat interface with the text 'Hi Shubha, How can I help?' and a button 'Set up an automation'. Below this, there are 'Recent chats' listed:

- Setting Up Data Quality Guardian Str... 19h ago
- Creating Internal Stage in Snowflak... 1mo ago
- Explore skills in your organization

At the bottom, there is a 'Logs' section with a search bar and a 'Filter' button. The status 'Running as ACCOUNTADMIN' is displayed. The footer includes 'Resource monitor: Off' and 'Feedback'.

04-rules-discovered.png



05-scan-score.png



08-guardrail-blocked.png

The screenshot shows the DQ Guardian interface. On the left, the 'Table to guard' is set to 'CUSTOMERS'. The 'Cortex model' is 'claude-sonnet-4-6'. The 'Scoring strictness' is set to 10.00. The 'Go to' menu shows 'Fix Studio' selected. The main panel displays a list of fixes. The first fix, 'Trim spaces in EMAIL - LOW - 91 rows - by TEMPLATE', is blocked by a guardrail. The error message states: 'Blocked by guardrail: Only UPDATE / DELETE / CREATE OR REPLACE TABLE ... AS SELECT on DQ_GUARDIAN.CLEAN.CUSTOMERS are allowed'. The fix SQL is: 'UPDATE DQ_GUARDIAN.CLEAN.CUSTOMERS SET "EMAIL" = TRIM("EMAIL") WHERE "EMAIL" IS NOT NULL AND "EMAIL" <> TRIM("EMAIL"); DROP TABLE *'. The interface also shows other fixes like 'Standardise spelling in STATE' and 'Standardise spelling in COUNTRY'.

09-before-after.png

The screenshot shows the DQ Guardian interface after applying fixes. The 'Table to guard' is still 'CUSTOMERS'. The 'Cortex model' is 'claude-sonnet-4-6'. The 'Scoring strictness' is 10.00. The main panel displays the 'Before' and 'After' results of applying fixes. The 'Before' score is 85.3, and the 'After' score is 87.0, with a change of +1.7. The 'Rows: source → sandbox' are 5,100 → 5,000. The interface shows a list of rules and their descriptions, along with a bar chart comparing the 'Before' and 'After' scores.

RULE_ID	SEVERITY	DESCRIPTION	FAILS
CUSTOMERS:DUPLICATE_KEY:CUSTOMER_ID	HIGH	CUSTOMER_ID must be unique (primary key)	
CUSTOMERS:DUPLICATE_ROW:*	MEDIUM	Exact duplicate rows	
CUSTOMERS:WHITESPACE:EMAIL	LOW	EMAIL has leading/trailing spaces	
CUSTOMERS:ALLOWED_VALUES:STATE	MEDIUM	STATE should use one consistent spelling (8 valid values)	
CUSTOMERS:REGEX:PHONE	MEDIUM	PHONE must be a 10-digit Indian mobile number	
CUSTOMERS:NOT_NULL:EMAIL	HIGH	EMAIL should not be null or blank	
CUSTOMERS:REGEX:PINCODE	MEDIUM	PINCODE must be a 6-digit PIN code	
CUSTOMERS:NOT_NULL:PHONE	HIGH	PHONE should not be null or blank	
CUSTOMERS:REGEX:EMAIL	HIGH	EMAIL must be a valid email address	
CUSTOMERS:ALLOWED_VALUES:COUNTRY	MEDIUM	COUNTRY should use one consistent spelling (1 valid values)	

10-audit-log.png

DQ Guardian

Source schema
RAW

Table to guard
CUSTOMERS

Cortex model
claudio-sonnet-4-6

☒ Use Cortex AI for fixes without a template

Scoring strictness
10.00

Go to

- Dashboard
- Scan
- Fix Studio
- CoCo Assist
- Audit

Source: DQ_GUARDIAN.RAW.CUSTOMERS Sandbox:
DQ_GUARDIAN.CLEAN.CUSTOMERS

Agentic Data Quality Guardian

Audit trail

Every sandbox change is recorded: who ran it, which role, what SQL, how many rows.

EVENT_TS	ACTION	ACTOR	ACTOR_ROLE	TARGET_FQN	ROWS_AFFECTED
2026-09-21 00:44:28	APPLIED	SHUBHAAI	ACCOUNTADMIN	DQ_GUARDIAN.CLEAN.CUSTOMERS	
2026-09-21 00:44:26	APPLIED	SHUBHAAI	ACCOUNTADMIN	DQ_GUARDIAN.CLEAN.CUSTOMERS	
2026-09-21 00:39:50	SANDBOX_CREATED	SHUBHAAI	ACCOUNTADMIN	DQ_GUARDIAN.CLEAN.CUSTOMERS	
2026-09-21 00:27:23	SANDBOX_CREATED	SHUBHAAI	ACCOUNTADMIN	DQ_GUARDIAN.CLEAN.CUSTOMERS	
2026-09-20 01:58:01	APPLIED	SHUBHAAI	ACCOUNTADMIN	DQ_GUARDIAN.CLEAN.CUSTOMERS	
2026-09-20 01:57:59	APPLIED	SHUBHAAI	ACCOUNTADMIN	DQ_GUARDIAN.CLEAN.CUSTOMERS	
2026-09-20 01:57:57	APPLIED	SHUBHAAI	ACCOUNTADMIN	DQ_GUARDIAN.CLEAN.CUSTOMERS	
2026-09-20 01:57:20	SANDBOX_CREATED	SHUBHAAI	ACCOUNTADMIN	DQ_GUARDIAN.CLEAN.CUSTOMERS	

Fix proposals

CREATED_AT	STATUS	RISK	GENERATED_BY	TITLE
2026-09-21 00:40:01	PROPOSED	MEDIUM	TEMPLATE	Standardise spelling in COUNTRY
2026-09-21 00:40:01	PROPOSED	MEDIUM	TEMPLATE	Quarantine invalid PIN codes (set to NULL)
2026-09-21 00:40:01	PROPOSED	MEDIUM	TEMPLATE	Quarantine out-of-range values in CREDIT_LIMIT (set to NULL)
2026-09-21 00:40:01	PROPOSED	MEDIUM	TEMPLATE	Standardise spelling in STATE
2026-09-21 00:40:01	PROPOSED	LOW	MANUAL	Needs review: Customer must be at least 18 years old at sign-up
2026-09-21 00:40:01	PROPOSED	MEDIUM	TEMPLATE	Quarantine out-of-range values in DATE_OF_BIRTH (set to NULL)
2026-09-21 00:40:01	PROPOSED	MEDIUM	TEMPLATE	Quarantine out-of-range values in SIGNUP_DATE (set to NULL)
2026-09-21 00:40:01	PROPOSED	MEDIUM	TEMPLATE	Normalise phone numbers, null the unrecoverable
2026-09-21 00:40:01	PROPOSED	MEDIUM	TEMPLATE	Standardise spelling in CUSTOMER_TIER
2026-09-21 00:40:01	APPLIED	LOW	TEMPLATE	Trim spaces in EMAIL

11-coco-assist-prompt.png

DQ Guardian

Source schema
RAW

Table to guard
CUSTOMERS

Cortex model
claude-sonnet-4-6

☒ Use Cortex AI for fixes without a template

Scoring strictness
10.00

Go to

- Dashboard
- Scan
- Fix Studio
- CoCo Assist
- Audit

Source: DQ_GUARDIAN.RAW.CUSTOMERS Sandbox:
DQ_GUARDIAN.CLEAN.CUSTOMERS

Agentic Data Quality Guardian

CoCo Assist

CoCo is Snowflake's AI coding agent (the icon at the bottom-right of Snowsight). This project uses CoCo at three points:

- Build time** - the app, rules and SQL were designed, reviewed and iterated with CoCo prompts (see `COCO_PROMPTS.md`).
- Run time - hand-off** - for issues the templates and Cortex cannot fix, this page builds a context-rich prompt you paste into CoCo. CoCo then inspects the table, previews the changes and applies them under your RBAC.
- Run time - Cortex** - fix SQL and the steward briefing come from Cortex LLM functions running inside Snowflake.

Hand-off prompt for CoCo

I am working in Snowsight with the database DQ_GUARDIAN. A data-quality scan of DQ_GUARDIAN.RAW.CUSTOMERS

- HIGH | NOT_NULL | EMAIL | 137 bad rows | EMAIL should not be null or blank
- HIGH | REGEX | EMAIL | 126 bad rows | EMAIL must be a valid email address
- HIGH | NOT_NULL | PHONE | 118 bad rows | PHONE should not be null or blank
- HIGH | DUPLICATE_KEY | CUSTOMER_ID | 166 bad rows | CUSTOMER_ID must be unique (primary key)
- MEDIUM | ALLOWED_VALUES | STATE | 531 bad rows | STATE should use one consistent spelling (8 valid value
- MEDIUM | REGEX | PHONE | 321 bad rows | PHONE must be a 10-digit Indian mobile number
- MEDIUM | ALLOWED_VALUES | COUNTRY | 146 bad rows | COUNTRY should use one consistent spelling (1 valid
- MEDIUM | REGEX | PINCODE | 136 bad rows | PINCODE must be a 6-digit PIN code
- MEDIUM | CUSTOM | SIGNUP_DATE | 122 bad rows | Customer must be at least 18 years old at sign-up
- MEDIUM | DUPLICATE_ROW | (table) | 166 bad rows | Exact duplicate rows
- MEDIUM | NOT_NULL | PINCODE | 85 bad rows | PINCODE should not be null or blank
- MEDIUM | ALLOWED_VALUES | CUSTOMER_TIER | 81 bad rows | CUSTOMER_TIER should use one consistent spelling
- MEDIUM | RANGE | CREDIT_LIMIT | 52 bad rows | CREDIT_LIMIT must not be negative
- MEDIUM | DATE_RULE | DATE_OF_BIRTH | 39 bad rows | DATE_OF_BIRTH must be a plausible birth date (not in
- MEDIUM | NOT_NULL | DATE_OF_BIRTH | 28 bad rows | DATE_OF_BIRTH should not be null or blank

Your task:

- Inspect DQ_GUARDIAN.RAW.CUSTOMERS and explain the likely root cause of each issue in one line.
- Write Snowflake SQL that fixes them ONLY on the sandbox clone DQ_GUARDIAN.CLEAN.CUSTOMERS (never on DC
- Before each change, show a SELECT that previews the affected rows and a COUNT.
- Prefer NULL over guessing when a value cannot be repaired with certainty.
- After running, re-check the failing conditions and report the before/after counts.
- List anything that needs a human decision instead of changing it.

12-coco-in-snowsight.png - 1 (CoCo Evidence)

In the database DQ_GUARDIAN, list every schema and table with its row count.
Then profile DQ_GUARDIAN.RAW.CUSTOMERS and DQ_GUARDIAN.RAW.ORDERS: which columns look like they have data-quality problems? Use SELECT statements only - do not modify anything.

Here's the full picture.

DQ_GUARDIAN — Schemas & Tables

Schema	Table	Row Count
CLEAN	CUSTOMERS	5,000
DQ	FIX_AUDIT_LOG	8
DQ	FIX_PROPOSALS	45
DQ	RULES	38
DQ	SCAN_RESULTS	180
DQ	SCAN_RUNS	5
RAW	CUSTOMERS	5,100
RAW	ORDERS	12,000

DQ_GUARDIAN CUSTOMERS FIX_AUDIT_LOG FIX_PROPOSALS RULES SCAN_RESULTS SCAN_RUNS
CUSTOMERS ORDERS

Data-Quality Problems in RAW.CUSTOMERS (5,100 rows)

Problem	Column(s)	Count	Details
Duplicate primary key	CUSTOMER_ID	100	100 IDs appear more than once
NULL emails	EMAIL	137	Missing contact info
Invalid email format	EMAIL	120	Don't match <code>%@%._%</code> pattern
Duplicate emails	EMAIL	98	Non-unique across customers
NULL phones	PHONE	118	Missing contact info

12-coco-in-snowsight.png - 2 (CoCo Evidence)

I am working in Snowsight with the database DQ_GUARDIAN. A data-quality scan of DQ_GUARDIAN.RAW.CUSTOMERS found:
- HIGH | NOT_NULL | EMAIL | 137 bad rows | EMAIL should not be null or blank
- HIGH | REGEX | EMAIL | 120 bad rows | EMAIL must be a valid email address...

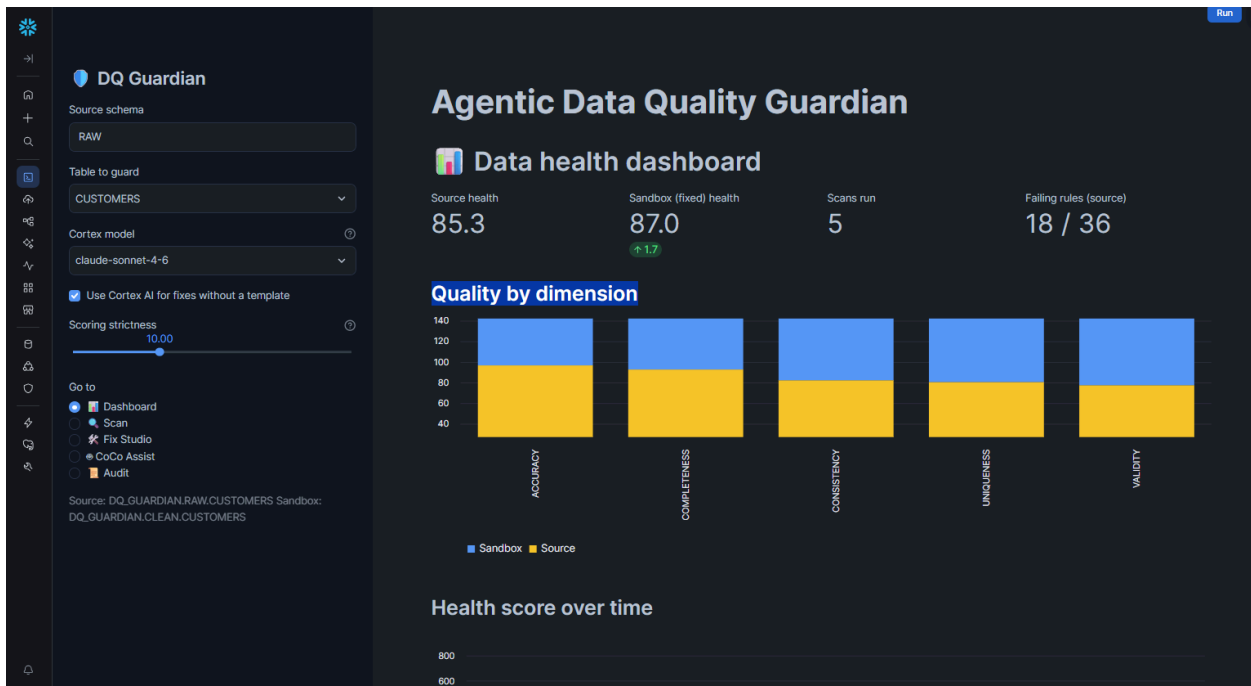
Here's the complete results.

Before / After Summary

Rule	Severity	Before (RAW)	After (CLEAN)	Status
Row count	—	5,100	5,000	-100 dupes removed
EMAIL format	HIGH	211	0	FIXED
CUSTOMER_ID duplicate	HIGH	100	0	FIXED
PHONE format	MEDIUM	321	0	FIXED
STATE values	MEDIUM	531	0	FIXED
COUNTRY values	MEDIUM	140	0	FIXED
PINCODE format	MEDIUM	130	0	FIXED
Duplicate rows	MEDIUM	100	0	FIXED
CUSTOMER_TIER values	MEDIUM	81	0	FIXED
CREDIT_LIMIT negative	MEDIUM	52	0	FIXED
DOB implausible range	MEDIUM	39	0	FIXED
EMAIL null	HIGH	137	135	HUMAN
PHONE null	HIGH	118	196	HUMAN
PINCODE null	MEDIUM	85	212	HUMAN
DOB null	MEDIUM	28	67	HUMAN
SIGNUP under-18	MEDIUM	122	122	HUMAN

11 of 15 rules fully fixed. 4 need human decisions.

13-dashboard-final.png



14-coco-security-review.png (CoCo Evidence)

Open the function `validate_fix_sql` in this Streamlit app. Explain what it allows and blocks. Then act as a red-teamer: list five ways an LLM-generated SQL string could still do damage, and for each one, say whether the current validator stops it and how to tighten it.

How `validate_fix_sql` works

The function is at `streamlit_app.py:470`. It takes a raw SQL string and the fully-qualified name of the `sandbox` table (`DQ_GUARDIAN.CLEAN.<table>`), and applies five layers of checks:

#	Check	Code	Effect
1	Empty check	<code>not sql or not sql.strip()</code>	Rejects blank in
2	Comment ban	<code>"--" in sql or "/*" in sql or "/" in sql</code>	Blocks SQL com
3	Statement count	<code>split_statements(sql) → split on ;</code>	Allows 1-6 state
4	Verb + target whitelist	Regex-matches each statement against three patterns: <code>UPDATE <target></code> , <code>DELETE FROM <target></code> , <code>CREATE OR REPLACE TABLE <target> AS SELECT</code>	Blocks any DDL,
5	Forbidden keyword blacklist	<code>FORBIDDEN_SQL</code> regex at line 467: <code>DROP</code> , <code>TRUNCATE</code> , <code>GRANT</code> , <code>REVOKE</code> , <code>ALTER</code> , <code>CALL</code> , <code>COPY</code> , <code>EXECUTE</code> , <code>UNDROP</code>	Blocks dangero

What it allows: Only UPDATE, DELETE, and CTAS targeting the single sandbox table. At most 6 statements. No comments, no DDL, no privilege changes.

What it blocks: Everything else — writes to any other table, DDL, privilege escalation, stored-procedure calls, COPY INTO, etc.

DQ Guardian

Source schema

RAW

Table to guard

CUSTOMERS ▼

Cortex model

claude-sonnet-4-0 ▼

☒ Use Cortex AI for fixes without a template

Scoring strictness

10.00

Go to

- ☐ Dashboard
- ☐ Scan
- ☐ Fix Studio
- ☒ CoCo Assist
- ☐ Audit

Source: DQ_GUARDIAN.RAW.CUSTOMERS Sanbox:
DQ_GUARDIAN.CLEAN.CUSTOMERS

Hand-off prompt for CoCo

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- MEDIUM | DUPLICATE_ROW | (cable) | 100 bad rows | Exact duplicate rows
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- MEDIUM | ALLOWED_VALUES | CUSTOMER_TIER | 81 bad rows | CUSTOMER_TIER s
- MEDIUM | RANGE | CREDIT_LIMIT | 52 bad rows | CREDIT_LIMIT must be
- MEDIUM | DATE_RULE | DATE_OF_BIRTH | 39 bad rows | DATE_OF_BIRTH must b
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4. Prefer NULL over guessing when a value cannot be repaired with certa
5. After running, re-check the failing conditions and report the before/s
6. List anything that needs a human decision instead of changing it.

Open CoCo in Snowsight (icon at the bottom-right), paste this prompt, and let it work on the sandbox clone.

Quick prompts to try in CoCo

Database Table Profiling and Da...

I am working in Snowsight with the database DQ_GUARDIAN. A data-quality scan of DQ_GUARDIAN.RAW.CUSTOMERS found:

- HIGH | NOT_NULL | EMAIL | 137 bad rows | EMAIL should not be null or blank
- HIGH | REGEX | EMAIL | 120 bad rows | EMAIL must be a valid email address...

id	rule	threshold
5	DUPLICATE_KEY CUSTOMER_ID (100)	50 (Each appear exactly twice — likely a double-load)
6	DUPLICATE_ROW (100)	Same 100 rows — the duplicates are exact copies
7	STATE bad values (539)	ALLCAPS (NARASIMHA), lowercase (narasimha), and types (Narasimha, oshiba)
8	COUNTRY bad values (140)	IND, INDIA, INDIA instead of canonical INDIA
9	CUSTOMER_TIER bad values (81)	Lowercase + types: brnze/brnze, gold/gld, silver/slier
10	PINCODE NULL (85)	
11	PINCODE bad format (103)	ABC-prefixed junk (48C3H), leading-zero 5-digit codes (403H), and 5-digit numbers (47396)
12	SIGNUP_DATE under-18 (122)	DOB + signup codes where age < 18 — some are children (age 5), some have future DOBs
13	DATE_OF_BIRTH implausible (39)	Future dates (2829-09-28) and impossibility old (1888-08-01)
14	DATE_OF_BIRTH NULL (28)	Missing values
15	CREDIT_LIMIT negative (52)	Sentinel value 5988 used instead of NULL

Now let me create the fresh sandbox clone and start fixing. I'll run the clone, then begin with deduplication since it must happen first (before row-level fixes).

Create fresh sandbox clone from RAW

CREATE or REPLACE TABLE DQ_SANDBOX.CLEAN.CUSTOMERS AS
SELECT * FROM DQ_GUARDIAN.RAW.CUSTOMERS;

Run this CREATE statement?

Add a follow up